G20 DELEGATES VISIT STATE-OF-THE-ART ASTRONOMY FACILITY IN THE KAROO







As South Africa celebrated Heritage Day on 24 September 2024, the nation's rich cultural heritage was complemented by a showcase of its scientific excellence when G20 delegates from over 30 countries gathered in the Northern Cape for a glimpse of the world's most sensitive radio telescope ever built, the MeerKAT and the Square Kilometre Array-Mid telescope. The tour to the site was a collaboration between the Department of Science, Technology and Innovation (DSTI) and the SKA Observatory (SKAO). The visit underscored the power of research to transform communities and South Africa's growing leadership in frontier science.

DSTI Director-a, Dr Mlungisi Cele, delivered the keynote on behalf of Minister Blade Nzimande, reminding guests that Africa is both the cradle of humanity and a rising force in global knowledge production.

"The SKA is more than a scientific endeavour, it's a driver for socio economic transformation, and a catalyst for developing the next generation of scientists, innovators, and engineers."

Through sustained government investment, the project has created more than 8 800 jobs in the region, channeled millions of rands into small local businesses, and trained over 300 artisans in Carnarvon. Ninety percent of the staff in the Northern Cape facilities are drawn from surrounding communities to ensure that world-class science leaves a tangible local footprint. "This is not just about telescopes," Dr Cele said. "It is about human potential."

Astronomy has long been a strategic priority for South Africa, deployed as a tool of science diplomacy and innovation. MeerKAT, already among the most powerful radio telescopes in the world and the world's most sensitive radio telescope, has produced more than 500 scientific publications. Its successor, SKA-Mid, is now under construction and promises to expand the country's role in probing cosmic origins.

Yet, as Dr Cele stressed, the ambition extends beyond discovery. The project continues to support the next generation of scientists and innovators through investments in education, including more than 1 700 bursaries with a 90% graduation rate, robotics programmes for schools, and early childhood development initiatives. Astro-tourism is being cultivated through a new visitor centre that links science to sustainable local enterprise.

Ultimately, science should remain grounded so that it can drive impact in communities. Prof. Jessica Dempsey, Director of the Netherlands Institute for Radio Astronomy, described the SKA as "probably the most diverse global experiment" ever attempted, not only in its science but in its model of cooperation.

This SKAO is a global enterprise that brings together countries from five continents, including Australia, Canada, China, France, Germany, India, Italy, the Netherlands, Portugal, South Africa, Spain, Sweden, Switzerland and the UK. Their roles span financing, operations, engineering, design and computing capacity, as well as infrastructure and science leadership. Together, these nations pool resources and innovation to drive breakthroughs in astronomy and technology while strengthening local and international capacity.

"Science is one of the last universal bridges we all share," said Prof. Dempsey, emphasising how collaboration across borders allows nations to pool wisdom and extend opportunities to young researchers worldwide. For Prof. Dempsey, the project's true return on investment is seen in the small and medium businesses exposed to cutting-edge technology, in young women inspired to pursue science, and in the commitment to diversity, equity and inclusion that ensures communities see themselves reflected in leadership. "If we close the gender gap in science," she said, "we would inject $5 trillion dollars into G20 GDP by 2030."

Local impact remains central. Dr Lindsay Magnus, Director of SKA-Mid, reflected on the responsibility to "be generators of information" rather than mere consumers. "South Africa must be an active player in the global technological and innovation landscape," he said. Dr Magnus believes the project's success will be measured as much in social legacy as in scientific breakthroughs.

"We intend to be in this community for the next 50 years," he said, highlighting ongoing support for schools and early development programmes. Plans are in place to expand the project and stretch operations far into the future. Dr Magnus emphasised that the project must continually uplift the communities hosting it.